

Retail Analytics Case Study:

From Data to Decision Support in Forecasting, Promotions & Store Performance

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Portfolio Purpose Explained



Solving Decision Support Problems

The portfolio demonstrates methods for tackling retail decision support challenges using realistic data and well-structured questions.



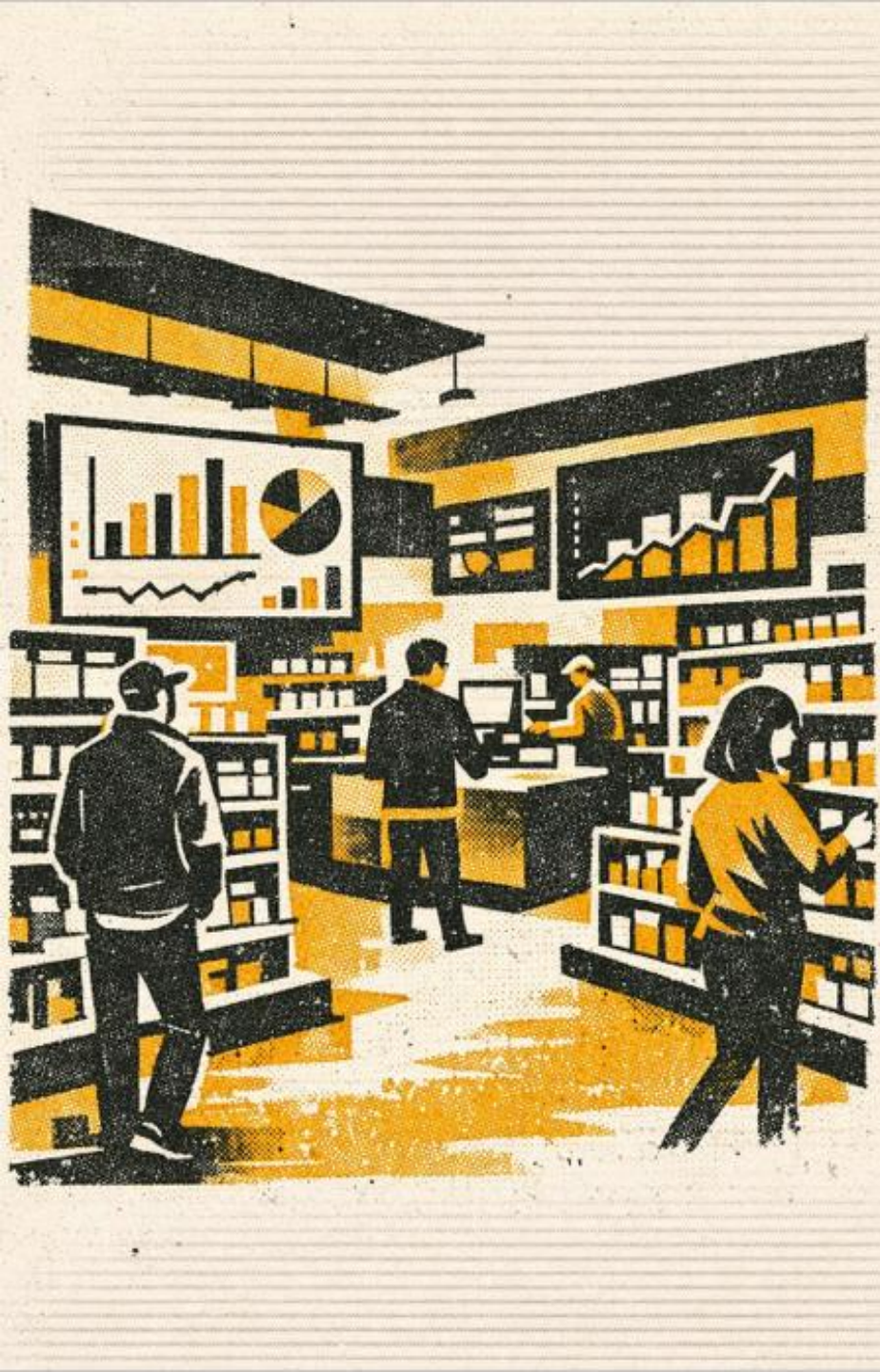
Real-world Project Scenarios

Projects are modeled on issues encountered by Senior Data Analysts in retail and real estate environments.



Turning Analysis into Insights

Technical analysis in the portfolio is translated into actionable business insights for executive decision making.



Agenda

- **Retail Data Landscape & Business Context**
- **Project Design and Data Framework**
- **Key Business Questions & Analytical Approach**
- **Insights & Decision Support Recommendations**
- **Future Enhancements & Next Steps**

Retail Data Landscape & Business Context

Key Retail Data Landscape - 1

A. Seasonal Sales Data

Retail businesses are heavily driven by:

- Winter (snow tires, heaters)
- Summer (BBQ, patio, camping)
- Holidays (Black Friday, Christmas)

Key questions:

- "Are we ready for seasonal demand?"
- "Did we overstock or understock?"

B. Product Category Performance

Just looking at "sales" is not enough:

- Automotive
- Tools
- Sports
- Outdoor / seasonal

Key question example:

- "Why are automotive sales down in Ontario but up in Alberta?"

Key Retail Data Landscape - 2

C. Promotions & Flyers

Retailers rely heavily on:

- Weekly flyers
- Discounts
- Loyalty programs

Key question example:

- “Did this flyer actually increase profit or just sales volume?”

D. Store-Level Performance

Canada is very diverse:

- Weather
- Income
- Population density

Key business focus areas:

- Sales per store
- Regional market differences
- Underperforming locations

Key Retail Data Landscape - 3

E. Real Estate & Space Optimization

Retailers must optimize limited physical space across stores:

- Shelf space is finite and high-value
- Product placement impacts sales performance
- Store size and layout vary across locations

Key questions:

- "Which products deserve more shelf space?"
- "Should we expand or shrink this store?"

F. Inventory & Stockouts (Hidden Revenue Killer)

Inventory management directly impacts both revenue and customer experience:

- Stockouts lead to lost sales and dissatisfied customers
- Overstock ties up capital and increases holding costs
- Demand variability makes inventory planning challenging

Key questions:

- "Where are we losing sales due to stockouts?"
- "Which products are consistently overstocked or understocked?"

Project Design and Data Framework

End-to-End Retail Analytics



Descriptive Analytics

Descriptive analytics helps businesses understand past retail performance by analyzing historical data and identifying patterns and trends.



Predictive Analytics

Predictive analytics uses models to forecast future retail outcomes, enabling companies to anticipate customer needs and market changes.

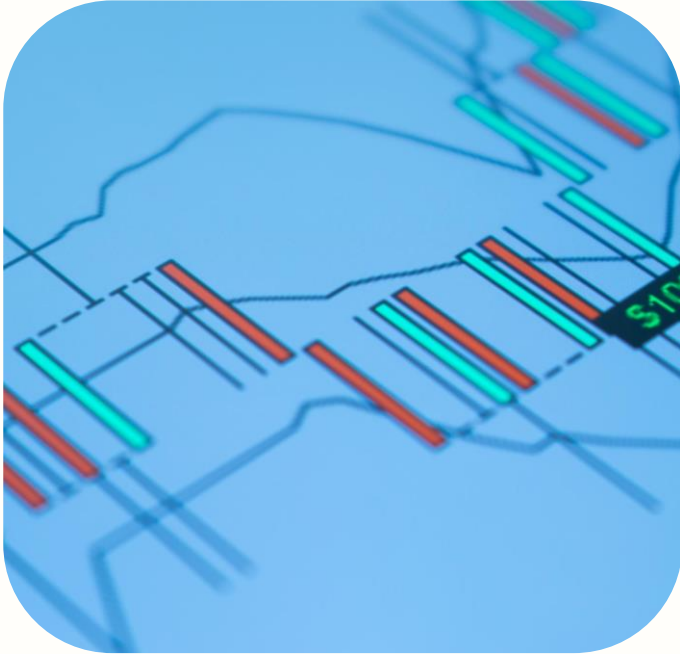


Prescriptive Analytics

Prescriptive analytics provides actionable recommendations, guiding businesses on optimal actions to drive strategic outcomes and improve performance.

Purpose-Driven Project Selection

Aligning analytics to business-critical retail questions



Key Idea:

- These projects follow a progression:
- Descriptive → Predictive → Prescriptive analytics
- Move from **understanding data** → **making decisions**

Demand Forecasting

Demand Forecasting uses data to predict sales trends, enabling retailers to optimize inventory and meet customer needs.

Promotion Effectiveness

Promotion Effectiveness analyzes campaign impacts, helping businesses refine marketing investments and maximize returns.

Store Performance Evaluation

Store Performance Evaluation examines operational metrics to improve efficiency and drive profitability across retail locations.

Scenario Simulator

Scenario Simulator models business strategies, supporting data-driven decisions for optimal business outcomes.

Synthetic Datasets & Data Model

Purpose and privacy

- Synthetic data simulates realistic retail business behavior
- Safe for experimentation and sharing

Data model structure

- Fact tables: sales, inventory, promotions
- Dimensions: products, stores, calendar, weather, demographics

Why it works

- Supports multi-dimensional analysis
- Enables forecasting, promo analysis, store benchmarking, and simulation

Realism for reliable testing

- Robust synthetic data reflects plausible seasonality, demand shifts, and category relationships, ensuring analytics behave as they would in production.
- This realism is critical for model testing and validation before deployment.

Data Architecture & Tables

The dataset was structured to mimic real enterprise retail systems

Core tables:

- products
- stores
- calendar
- weather
- demographics
- promotions
- inventory
- daily_sales

Additional:

- store-product assortment matrix

Key Idea:

Multi-table relational structure similar to real-world data warehouse

Key Business Questions & Analytical Approach

Project 1: Business Problem

Demand Forecasting

Adjust inventory targets

This project addresses:

“How can we model and forecast demand under strong seasonality and external drivers?”

Question:

How can we accurately forecast demand for seasonal retail products?

Challenges:

strong seasonality
weather impact
demand spikes

Project 1: Model & Approach

This Project was approached in multiple stages:

1. Baseline (simple time-based / moving average logic)

- Used basic aggregation and rolling averages
- Purpose: establish a benchmark and understand seasonality patterns
- Limitation: could not capture complex trends or external drivers

2. Time Series Model – Prophet

- Chosen because it handles:
 - seasonality (weekly, yearly)
 - trend changes
- More interpretable for business stakeholders
- Improvement: captured overall demand patterns well

3. Prophet + External Features (Weather)

- Added:
 - temperature
 - snowfall
- Reason: retail demand (especially for seasonal retail products) is heavily weather-driven
- Improvement: better alignment with real-world demand spikes

4. Hybrid Model (Time Series + ML logic)

- Used time series for baseline + additional logic to adjust spikes (e.g., promotions, extreme weather)
- Reason: pure time series tends to **underestimate peaks**

Project 1: Results & Insights

Model Insights:

- Successfully captured strong seasonal trends
- Weather significantly improved forecast realism
- Identified key limitation:

Peak demand underestimation → risk of stockouts

Business value:

Improves inventory planning and highlights when additional safety stock is needed

Technical takeaway:

Need for hybrid or feature-rich models (e.g., gradient boosting, sequence models like LSTM) for peak capture

Project 1: Example Results

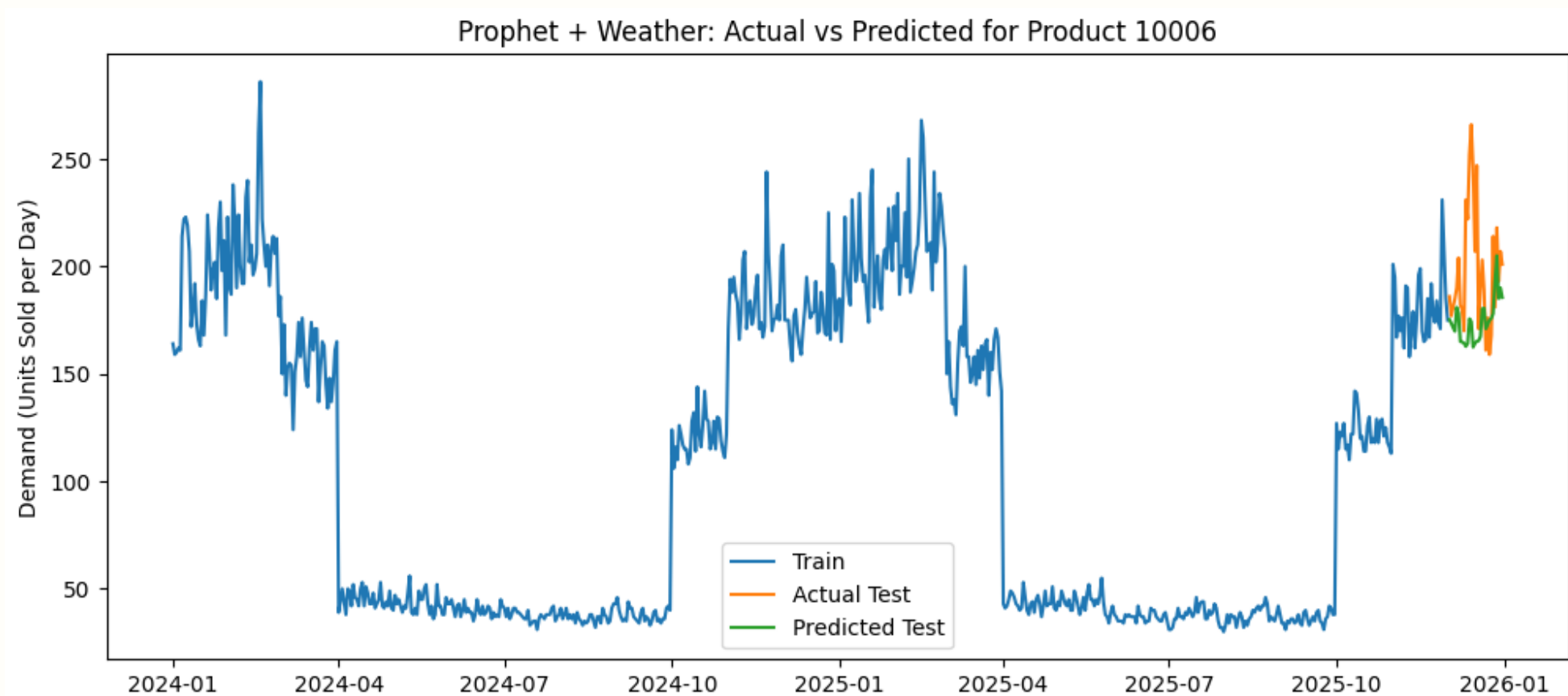
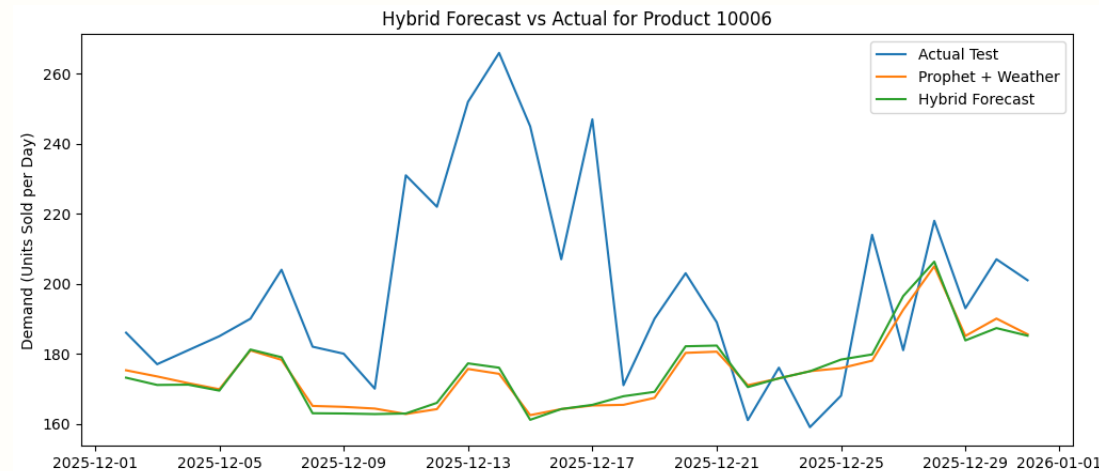
Metrics: (Example: Product 10006, Prediction for next 90 days)

Random Forest MAE: 55.82

Prophet MAE: 27.38

Prophet + Weather Test MAE: 27.0

Hybrid Test MAE: 27.17



Project 2: Business Problem

Promotion Effectiveness

Optimize promotion timing, and improves ROI

This project addresses:

“What is the causal and financial impact of promotions?”

Question:

Do promotions increase profitable growth?

Challenges:

non-random promotion assignment

distinguishing price-driven vs volume-driven (demand) effects

lack of true baseline (no-promo scenario)

Project 2: Analytical Approach

Prioritized causal analysis and business insight over predictive modeling:

- Built a **transaction-level analytical dataset**:
 - joined sales + product + promotion data
- Created features:
 - has_promo
 - promo_type
 - margin_rate
- Used **descriptive analytics and aggregation**:
 - groupby aggregation
 - pivot analysis
 - segmentation
- Compared:
 - promo vs non-promo
 - promo types (BOGO, loyalty, % off, etc.)
 - category-level response

Project 2: Results & Insights

Results & Insights:

- Promotions increased units sold by ~31%
- But reduced margin significantly (~42% → ~31%)
- Strong variation by:
 - **promo type** (BOGO = high volume, low margin)
 - **category** (Outdoor Living ~109% uplift vs Automotive low response)

Business value:

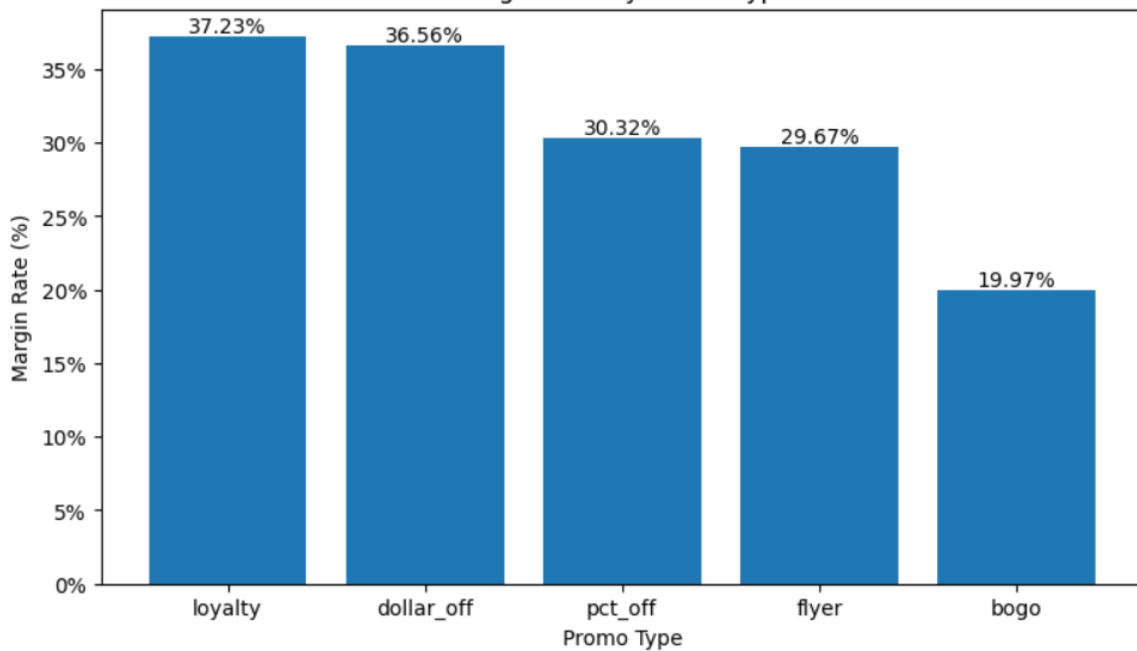
- Promotions should be targeted, not applied broadly
- Enables better balance between revenue growth and profitability

Technical Insight:

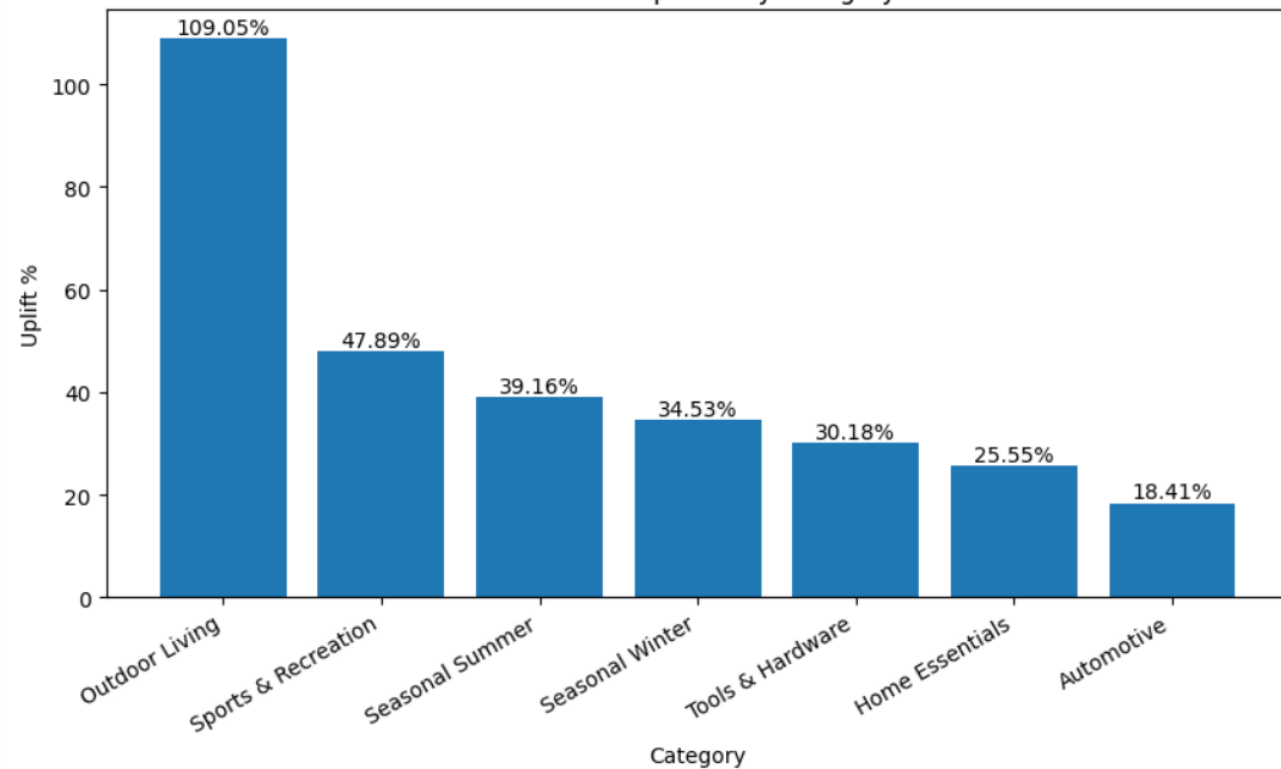
- Potential bias due to non-random promotion assignment
- Future improvement:
 - Use causal inference methods (e.g., matched comparison)
 - Estimate true incremental uplift from promotions

Project 2: Example Results

Margin Rate by Promo Type



Promotion Uplift % by Category



Project 3: Business Problem

Store Performance

Identifies high-potential locations

This project addresses:

“What features explain variance in store-level performance?”

Question:

Why do some stores perform better than others, and how can we optimize store strategy?

Challenges:

store heterogeneity (location, size, demographics)
limited control over external factors
difficulty isolating true performance drivers

Project 3: Analytical Approach

No heavy ML model was used because the objective was interpretability and decision support

- **Analytical Approach & Methods :**

- segmentation analysis
- statistical comparisons
- feature engineering (demographics, competition, store type)

- **Key Metrics Introduced:**

- sales per square foot → efficiency metric
- performance segmentation → high / medium / low stores

- **Feature Engineering**

Constructed store-level dataset with:

- Demographic features:
 - median income
 - household composition
- Competition features:
 - competitor_store_count
- Store characteristics:
 - store_type (urban/suburban/rural)

Project 3: Results & Insights

Model Insights:

- Suburban stores outperform urban and rural
- High-income areas → higher sales
- High competition → lower performance
- Promotion effectiveness varies by store

Business value:

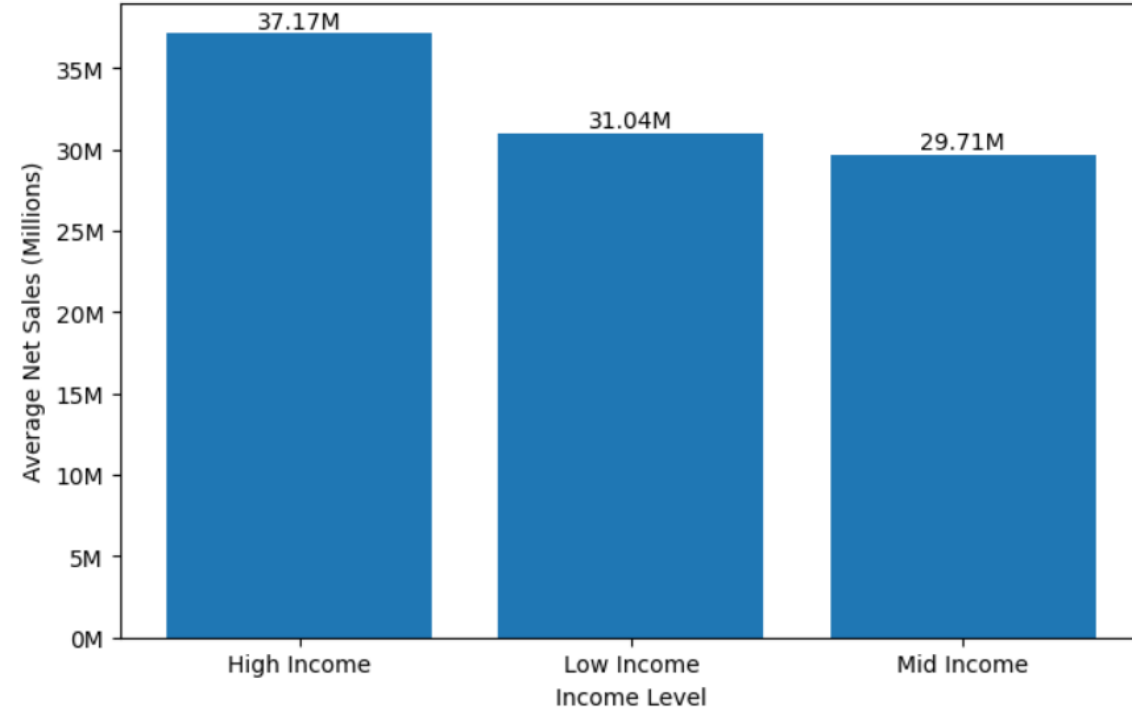
- Supports location strategy & real estate decisions
- Enables localized pricing, promotion, and inventory strategies

Technical Insight:

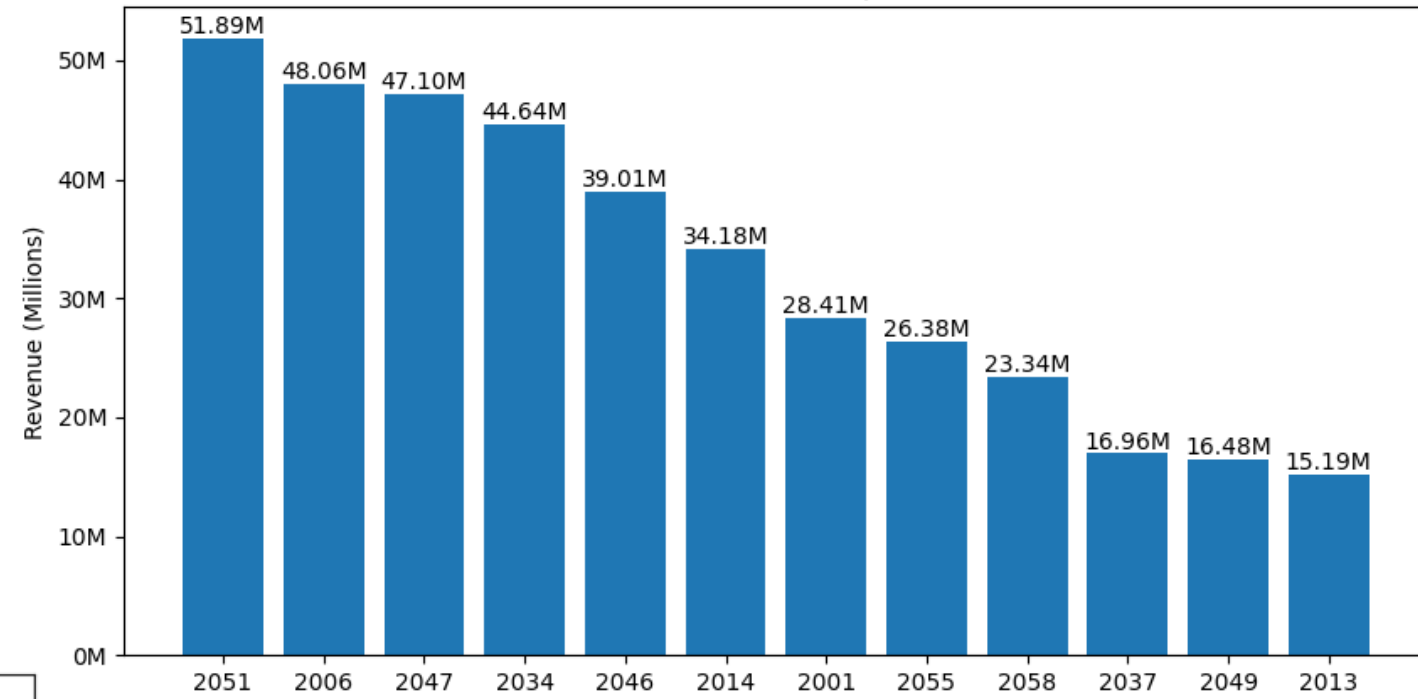
- Stores operate under different conditions
- Advanced models can:
 - adjust for these differences
 - isolate true store performance
- Regression models to quantify impact of (Potential Extension):
 - income
 - competition
 - store type
- Enables identification of:
 - over-performing stores (best practices)
 - under-performing stores (improvement opportunities)

Project 3: Example Results

Average Net Sales by Income Level



Store Revenue Comparison



Project 4: Business Problem

Scenario Simulator

Reduce uncertainty in inventory, pricing, and promotion decisions

This project addresses:

How can we simulate and evaluate decisions before execution?

Question:

What will happen if we change pricing, promotions, or inventory before we actually do it?

Challenges:

simplifying real-world complexity into rules
assumptions in scenario design
trade-offs between accuracy and interpretability

Project 4: Model & Approach

Built a scenario simulation engine (rule-based + data-driven)

Inputs:

- Discount changes
- Category selection
- Store selection

Uses outputs from:

- Project 1 (demand patterns)
- Project 2 (promo uplift)
- Project 3 (store differences)

AI Layer (Extension)

- Built a **lightweight AI-style assistant interface**
- Converts business questions → simulations → recommendations

This is **prescriptive analytics**, not just descriptive or predictive.

Project 4: Results & Insights

Insights:

- Demonstrates trade-offs:
 - some categories → high demand uplift but margin loss
 - others → minimal benefit from promotions
- Enables comparison across scenarios

Business value:

- Supports **decision-making before execution**
- Helps answer:
 - "Should we increase discounts?"
 - "Where should we apply promotions?"
- Moves organization toward **data-driven and proactive strategy**

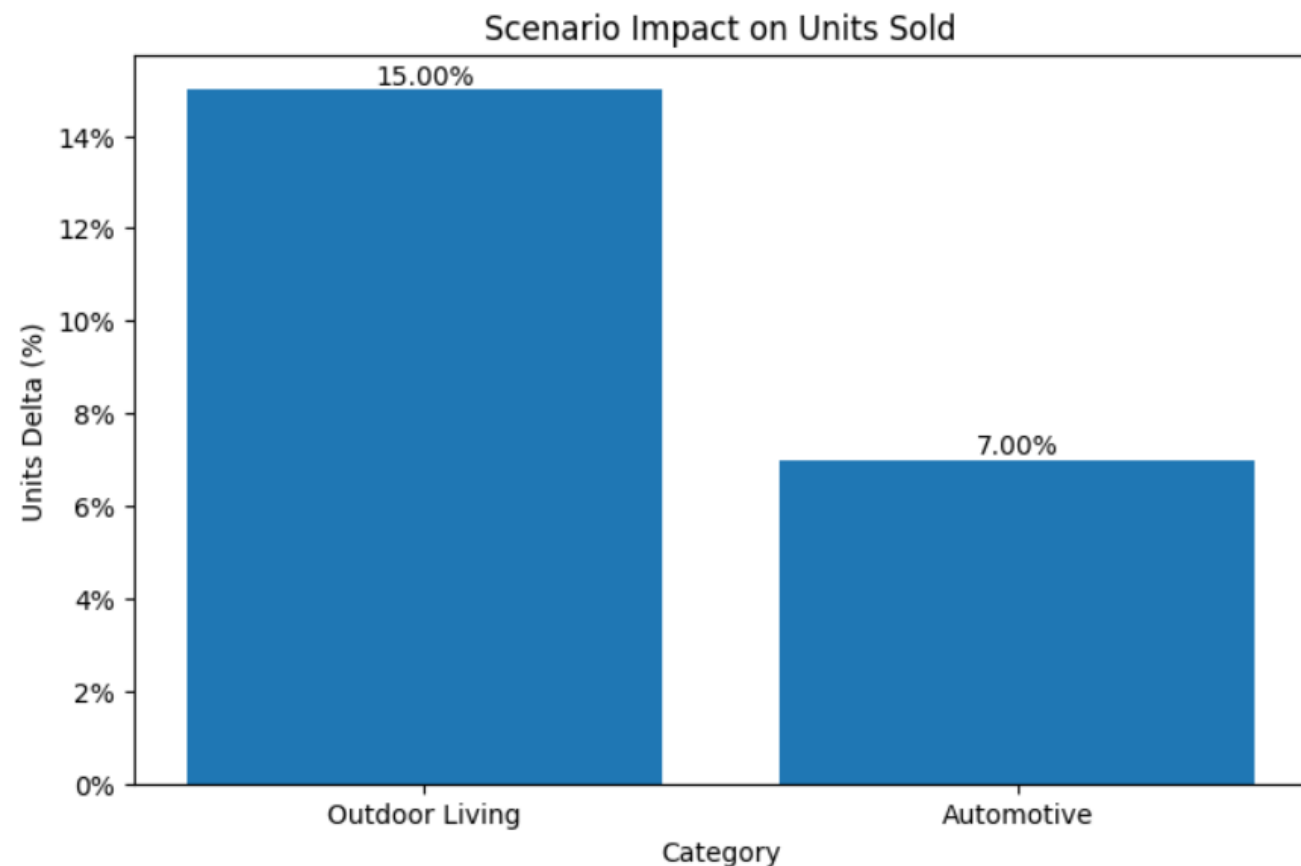
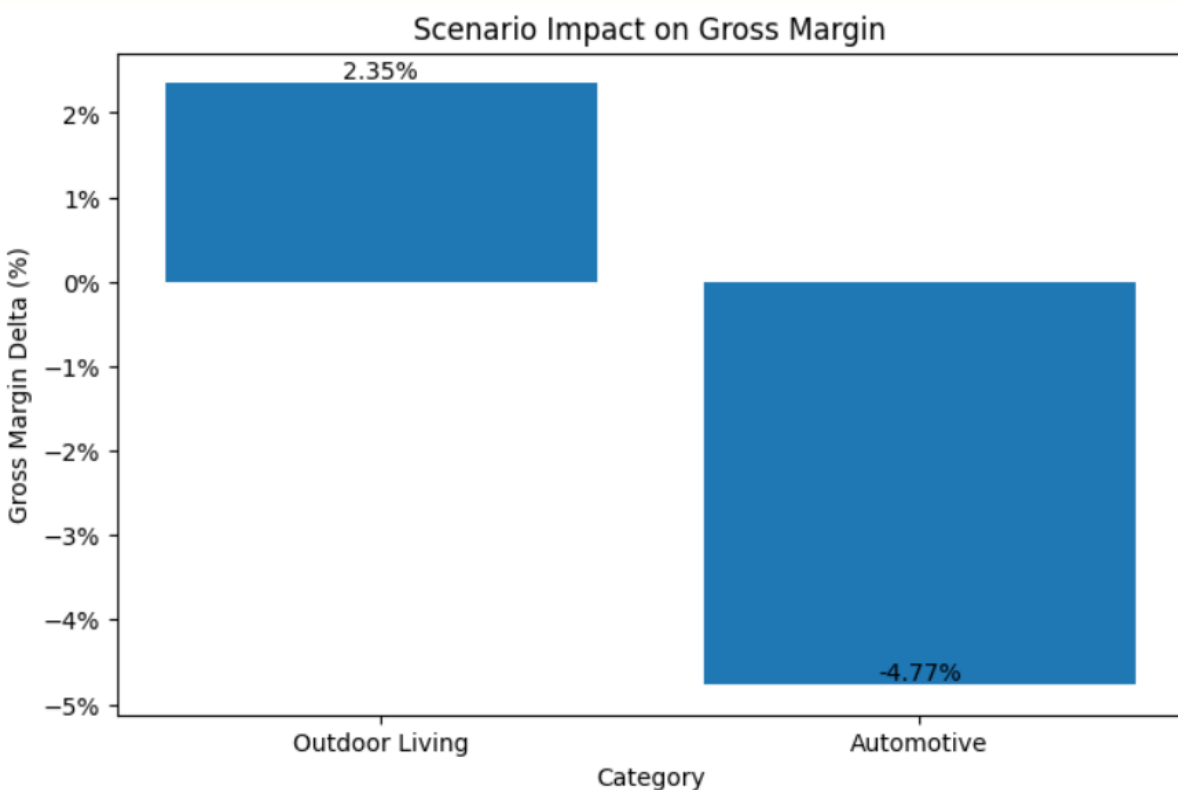
Technical Insight:

- Current system:
 - deterministic + rule-based
- Extensions:
 - probabilistic simulation
 - reinforcement learning for policy optimization
 - integration with LLM + RAG

Project 4: Example Results

Scenario A: Outdoor Living discount increase

Scenario B: Automotive discount increase



```
questions = [  
    "What happens if we increase discount from 10% to 20% for Outdoor Living?",  
    "What happens if we increase discount from 10% to 20% for Automotive?",  
    "What happens if we increase discount for Seasonal Winter?"  
]
```

Insights & Decision Support Recommendations

Projects Supporting Executive Decisions

Each analytics initiative reduces uncertainty and enables faster, and confident decisions aligned to corporate goals.



Demand Planning

Analytics projects forecast sales trends and inventory needs, enabling executives to optimize stock levels and reduce overstock risk.



Promotion Budgeting

Data-driven insights help allocate promotional budgets more effectively, maximizing ROI and supporting strategic marketing decisions.



Store Investments

Project outputs identify high-potential locations and investment opportunities, guiding capital allocation for store expansion or upgrades.



Pricing Strategies

Analytics inform dynamic pricing models, allowing executives to adjust prices in response to market changes and competitor actions.

Key Business Findings (Store Performance & Promotion Effectiveness)

“What is happening?” → observations

1. Suburban stores are strongest

Suburban → highest revenue & units → ~42M avg net sales

Urban → lowest → ~20M

2. Income level matters

High Income → ~37M

Low/Mid → ~30M

3. Competition hurts performance

Low competition → highest sales → ~35M

High competition → lowest → ~27M

4. Efficiency varies significantly across stores

Sales per sqft:

Top store → 1162\$/sqft (Store 2013)

Others → lower 336\$/sqft (Store 2049)

5. Promotions don't work equally across stores

Example:

- Some stores → +29% lift
- Others → negative lift

Example:

- Store 2006 → -4.4%
- Store 2058 → ~0%

6. Stores can be segmented based on performance

Example:

- High performers → 2013, 2058, 2037, 2047
- Low performers → 2034, 2014, 2055, 2049

Example:

- High revenue → 2051, 2006, 2047, 2034
- Low revenue → 2058, 2037, 2049, 2013

Key Business Insights (Store Performance & Promotion Effectiveness)

“What is happening?” → insights

Insight 1 – Performance is location-driven

- Income → Demographics matter

“Stores in higher-income areas tend to generate higher sales, suggesting purchasing power plays a key role.”

- Competition → Competition impacts performance

“Stores facing higher competition show lower average revenue, indicating market saturation effects.”

- Suburban setting

Insight 2 – One-size-fits-all strategy is WRONG

- Promotions work differently per store → Promo dependence differs

“Some stores rely heavily on promotions to drive sales, while others maintain stable performance without heavy discounting.”

- Categories behave differently

- Demand drivers differ

store_id	Top 1	Top 2
2034	Home Essentials (27.0% margin, 2.7% uplift)	Tools & Hardware (35.5% margin, 2.5% uplift)
2037	Tools & Hardware (33.5% margin, 1.3% uplift)	Automotive (31.6% margin, 0.2% uplift)
2046	Tools & Hardware (36.7% margin, 1.9% uplift)	Outdoor Living (28.3% margin, 1.0% uplift)
2047	Home Essentials (26.6% margin, 2.8% uplift)	Tools & Hardware (36.3% margin, 2.6% uplift)
2055	Tools & Hardware (33.7% margin, 2.0% uplift)	Home Essentials (28.6% margin, 1.6% uplift)

Insight 3 – Efficiency matters, not just revenue

- High revenue ≠ best store

“Sales per square foot reveals that some lower-revenue stores are actually more efficient.”

Executive Recommendations Summary (Store Performance & Promotion Effectiveness)

“What should we do?” → decisions

1. Localized strategy

“Adopt store-level strategies rather than uniform pricing, promotion, and assortment policies.”

- high promo-dependence stores → optimize pricing (Store 2047,2046,2034,2037)
- low dependence → preserve margin (Store 2006,2058,2014,2001)

“Stores with low or negative promo uplift should reduce discounting and focus on margin preservation.”

2. Fix underperforming stores

For low performers (Store 2034, 2014, 2055, 2049):

- adjust product mix
- optimize inventory
- re-evaluate promotion strategy

3. Use top stores as benchmark (Store 2013,2058,2037,2047)

“Use high-performing stores to define best practices in layout, assortment, and pricing.”

Executive Recommendations Summary (Store Performance & Promotion Effectiveness)

“What should we do?” → decisions

4. Reduce ineffective promotions and Increase effective promotions

→ Target promotions by category

- use aggressive promos for:
 - Outdoor Living → ~109% uplift
 - Seasonal categories → ~40% uplift
- reduce promos for:
 - Automotive → ~18% uplift

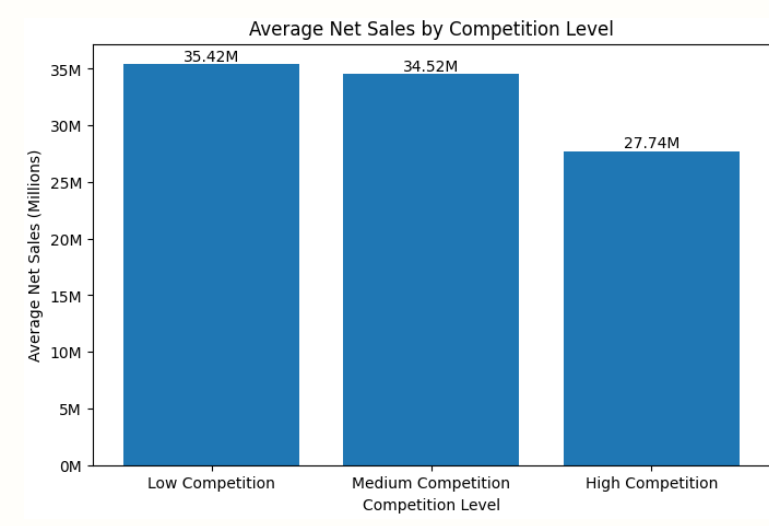
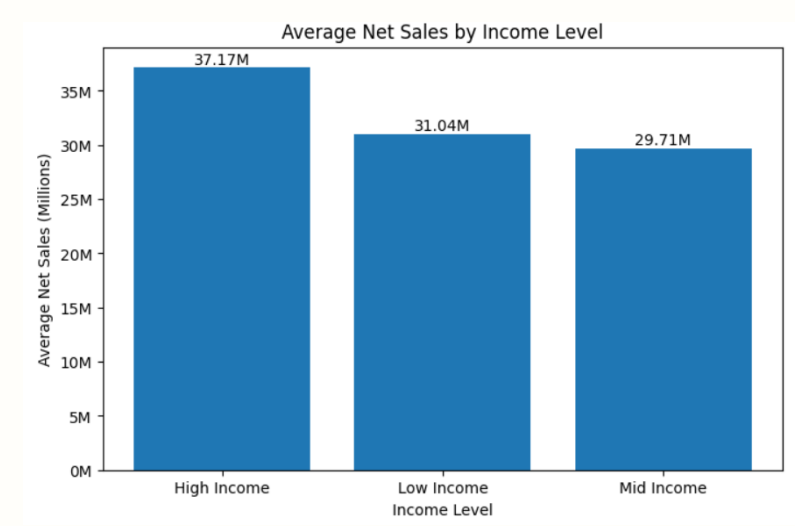
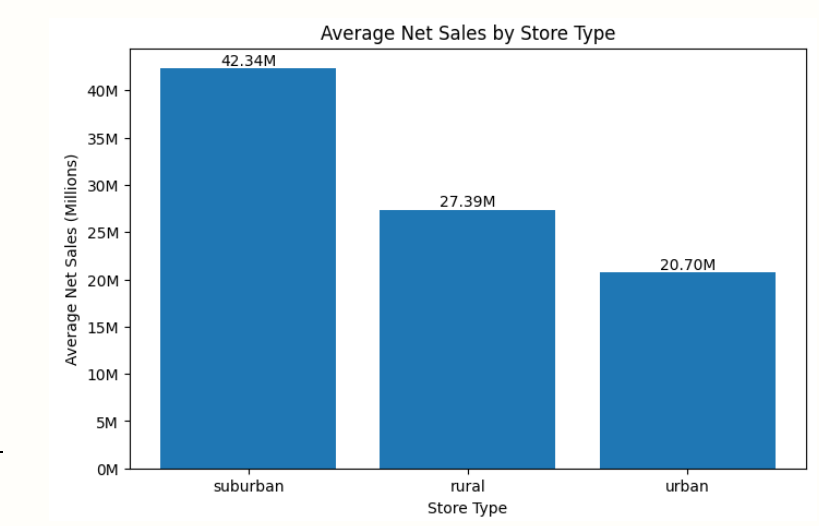
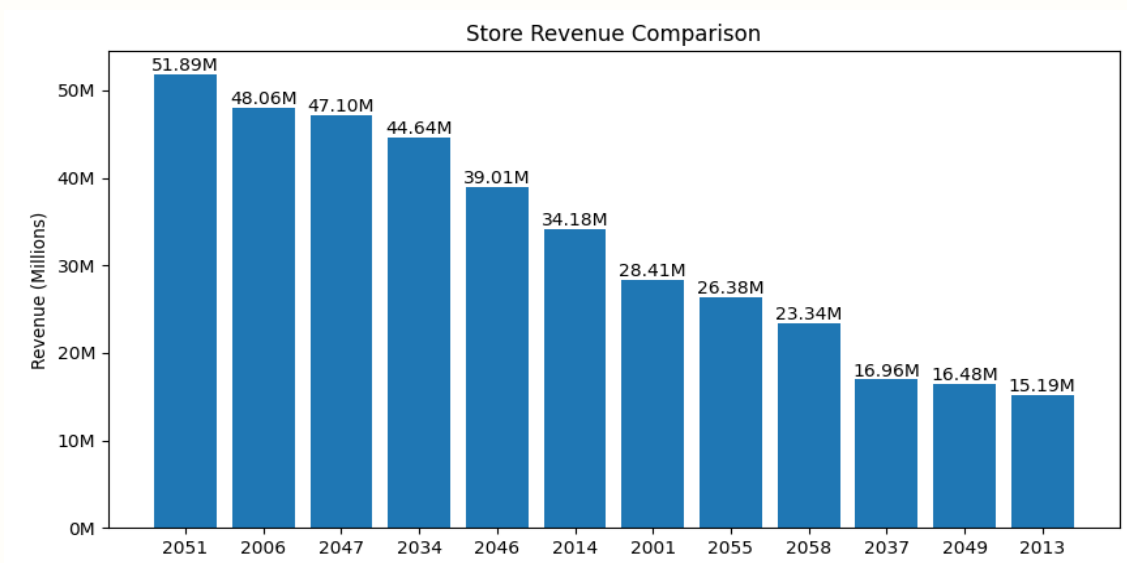
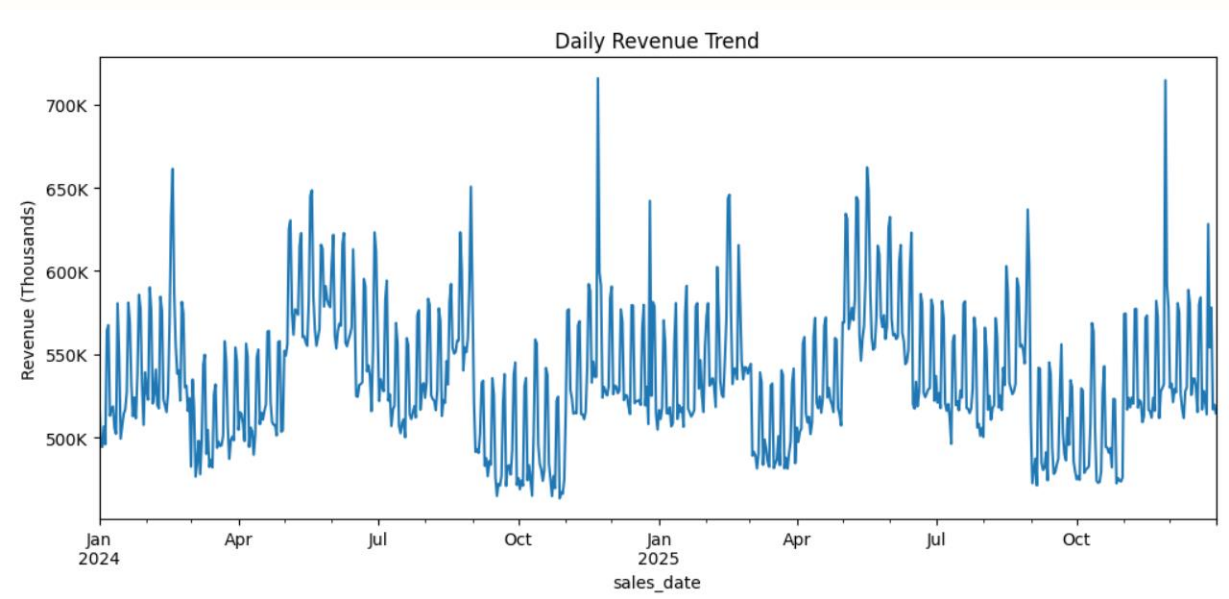
→ Optimize promo types

- reduce use of:
 - BOGO (low margin) → ~20%
- increase use of:
 - loyalty / dollar_off → ~37%

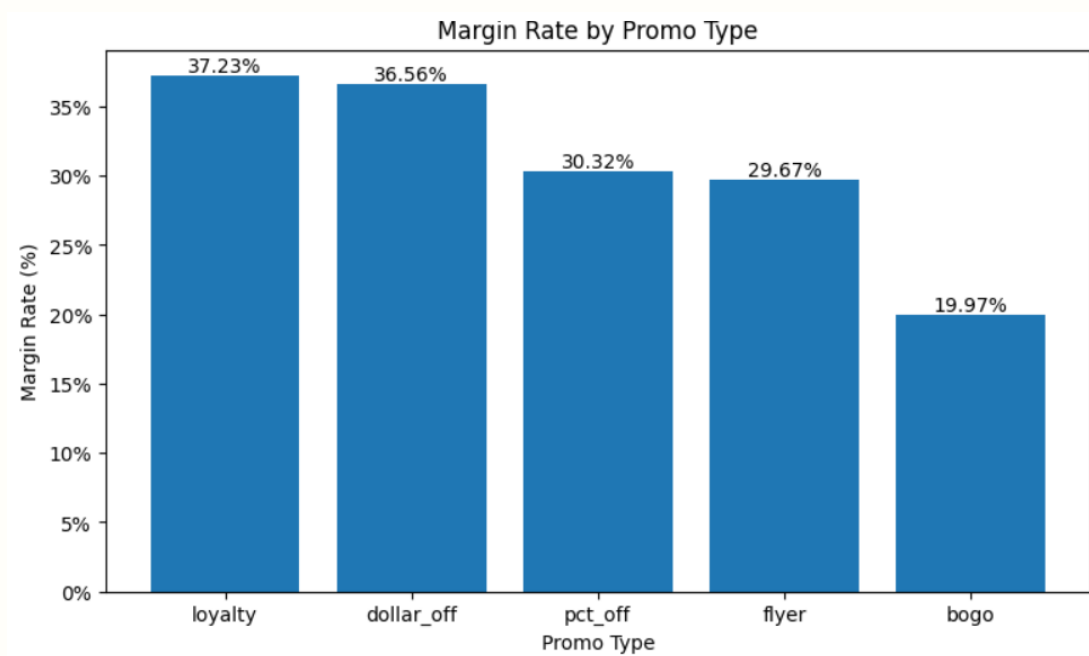
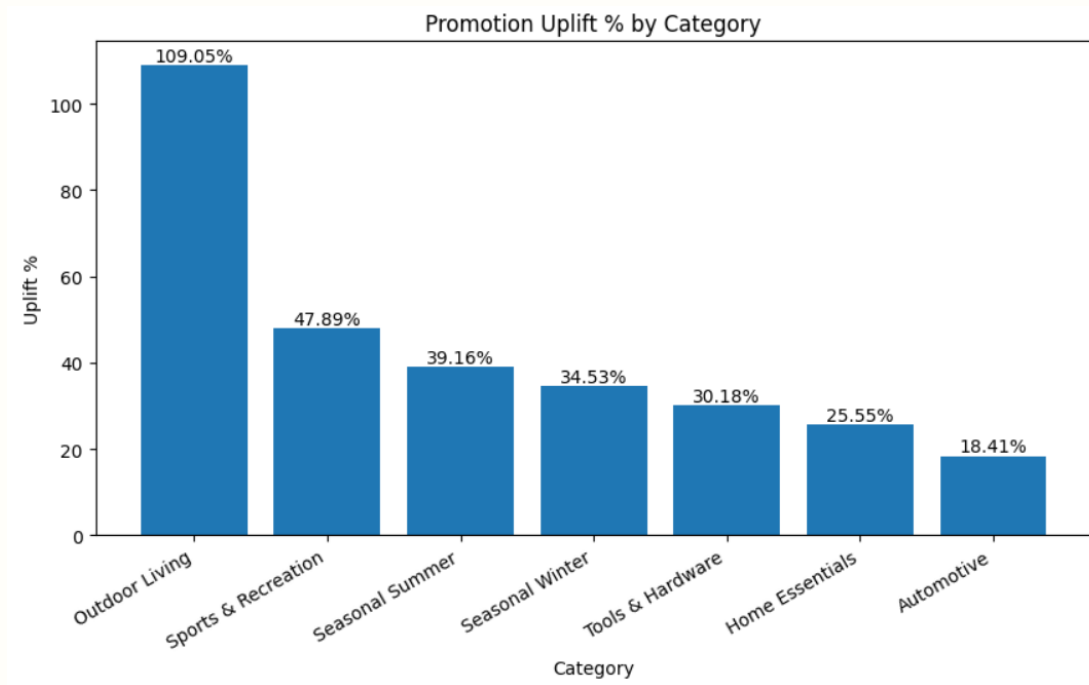
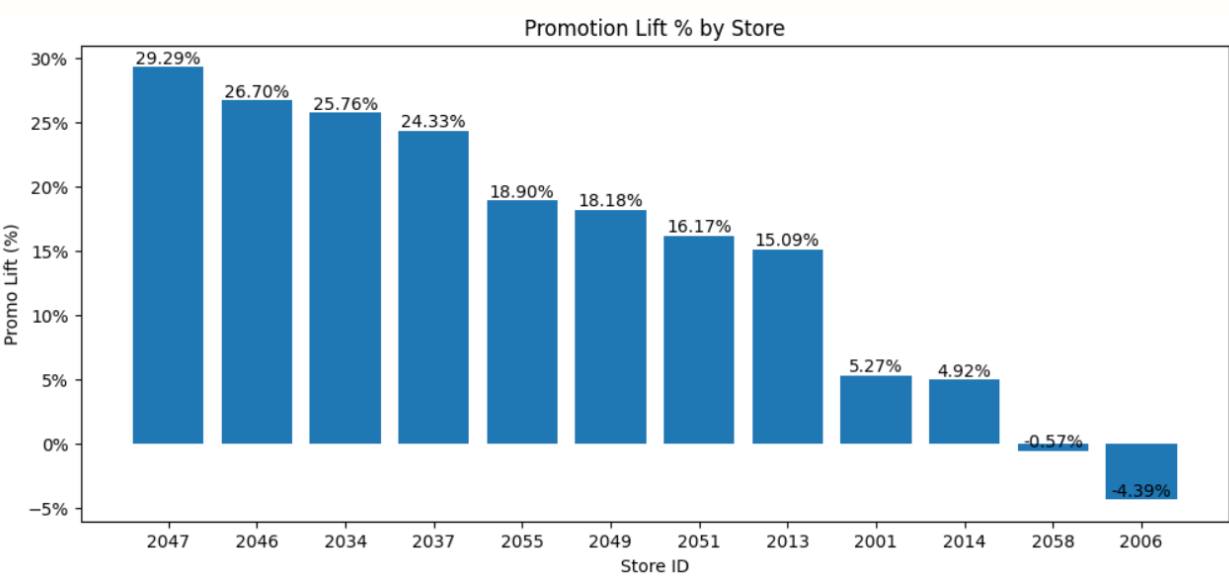
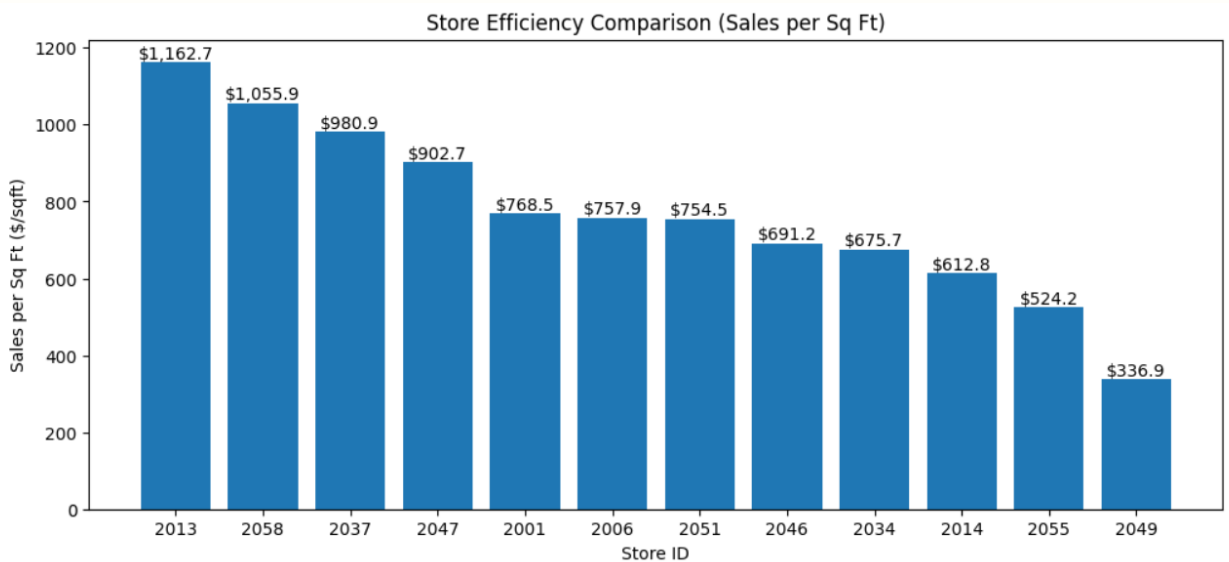
5. Real estate / expansion strategy

“Future store expansion should prioritize suburban, high-income areas with lower competition.”

Core KPI Dashboard Examples



Core KPI Dashboard Examples



Future Enhancements & Next Steps

Future Enhancements & Next Steps

Opportunities for Improvement and Roadmap for Advanced Analytics

- **Interactive Executive Dashboards (Power BI)**

Develop dynamic dashboards to allow executives to explore KPIs, scenarios, and trends in real time, enabling faster and more informed decision-making.

- **Advanced Forecasting Models**

Extend demand forecasting using models such as ARIMA, LSTM, or gradient boosting (e.g., XGBoost) to better capture non-linear patterns and improve peak demand prediction.

- **Hyperparameter Optimization & Model Tuning**

Improve model performance through systematic tuning (cross-validation, grid/random search), ensuring more accurate and robust predictions.

- **Real-Time Data Integration**

Incorporate near real-time data pipelines (e.g., streaming sales, inventory updates) to support faster and more responsive decision-making.

- **Generative AI Integration (LLM + RAG)**

Enable natural language interaction with data, allowing executives to ask business questions and receive automated insights and recommendations.

- **Scenario Optimization & Decision Automation**

Extend the scenario simulator toward optimization (e.g., identifying the best pricing or promotion strategy automatically) and potential agent-based decision systems.

